Predicting Term Deposit Subscription

A Machine Learning Approach for Bank Marketing Decision-Making

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- Dataset: Bank Marketing Dataset (UCI)

BACKGROUND: INDUSTRY CONTEXT

Retail Banking Marketing Context

Retail banks increasingly face diminishing returns from traditional mass outbound marketing campaigns. Telemarketing activities incur high operational costs while generating persistently low response rates, and are further constrained by customer fatigue, reputational concerns, and regulatory requirements. As a result, indiscriminate contact of the entire customer base is inefficient and economically unsustainable.

In this context, marketing effectiveness depends critically on the ability to prioritise customers based on their likelihood of response. Prior empirical evidence shows that analytics-driven targeting substantially improves campaign performance by reallocating resources toward high-propensity customers, thereby increasing conversion rates while reducing unnecessary contact costs (Moro et al., 2014).

ANALYTICS & DATA MINING CONTEXT

From an analytics perspective, customer response prediction is naturally formulated as a binary classification problem, where the objective is to estimate the probability of subscription for each customer. Logistic regression, decision trees, and ensemble models are widely applied in this setting, each offering different trade-offs between interpretability, flexibility, and predictive power.

From a data mining standpoint, this study follows a structured workflow to extract multiple types of knowledge from data. Descriptive knowledge is used to assess data quality and distributional properties, predictive knowledge is derived to estimate subscription likelihood, and segmentation knowledge is obtained through tree-based models to support customer profiling and managerial interpretation (Kuhn & Johnson, 2013).

BUSINESS PROBLEM

The business objective of this project is to increase term deposit subscriptions while simultaneously reducing unnecessary marketing expenditure. The response variable indicates whether a customer subscribes to a term deposit (“yes” or “no”). Descriptive analysis reveals that only approximately 11% of customers subscribe, indicating a severely imbalanced outcome and highlighting the inefficiency of mass marketing strategies. Given the high marginal cost of outbound calls, even small improvements in customer prioritisation can generate substantial operational and financial benefits, making accurate ranking of customers a critical managerial challenge.

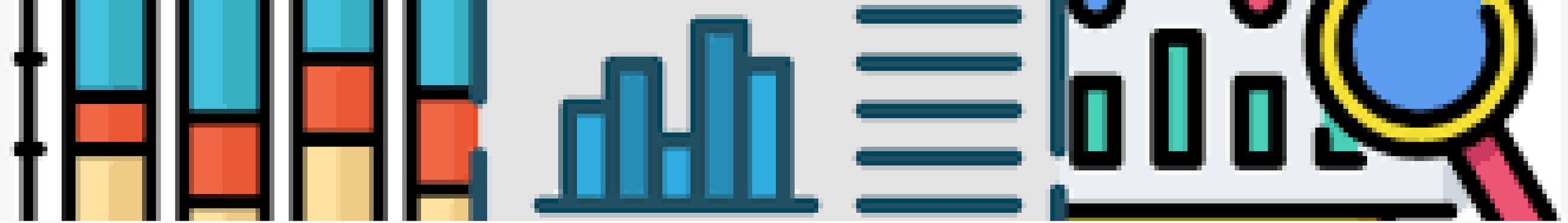
BUSINESS OBJECTIVE & KPI

Because the response variable is a binary asymmetric attribute, overall classification accuracy is a misleading performance metric. The business cost of failing to contact a potential subscriber is substantially higher than the cost of contacting an uninterested customer.
Consequently, model evaluation focuses on metrics that reflect ranking quality and business effectiveness rather than error minimization alone. Recall is used to measure the model's ability to capture subscribers, lift evaluates improvement over random targeting, and AUC assesses overall ranking discrimination across thresholds. These metrics are well suited to marketing analytics applications involving imbalanced outcomes (Provost & Fawcett, 2013; Verbeke et al., 2012).

PROPOSED SOLUTION

Three model families are considered to address the business problem. Logistic regression is selected as the primary deployment model due to its interpretability, stable probability estimates, and suitability for regulated decision-making environments.

A decision tree model is employed to support insight discovery and customer segmentation by revealing non-linear interaction structures in an interpretable form. In addition, a random forest model is implemented as a benchmark to assess the upper bound of predictive performance, rather than as a deployment candidate, consistent with best practices in regulated business analytics (Breiman, 2001).



DATA & VARIABLES

The analysis is based on the Bank Marketing Dataset from the UCI Machine Learning Repository, comprising 41,188 customer observations collected from historical telemarketing campaigns (UCI ML Repository, 2014).

The dataset includes nominal categorical variables such as job, education, and marital status; discrete behavioural variables such as campaign and previous contact history; and continuous macroeconomic indicators capturing broader economic conditions. The response variable is a binary asymmetric attribute indicating term deposit subscription. The presence of mixed data types, class imbalance, and skewed behavioural distributions motivates careful data preparation prior to model development.

Data Quality and Distribution Diagnostics

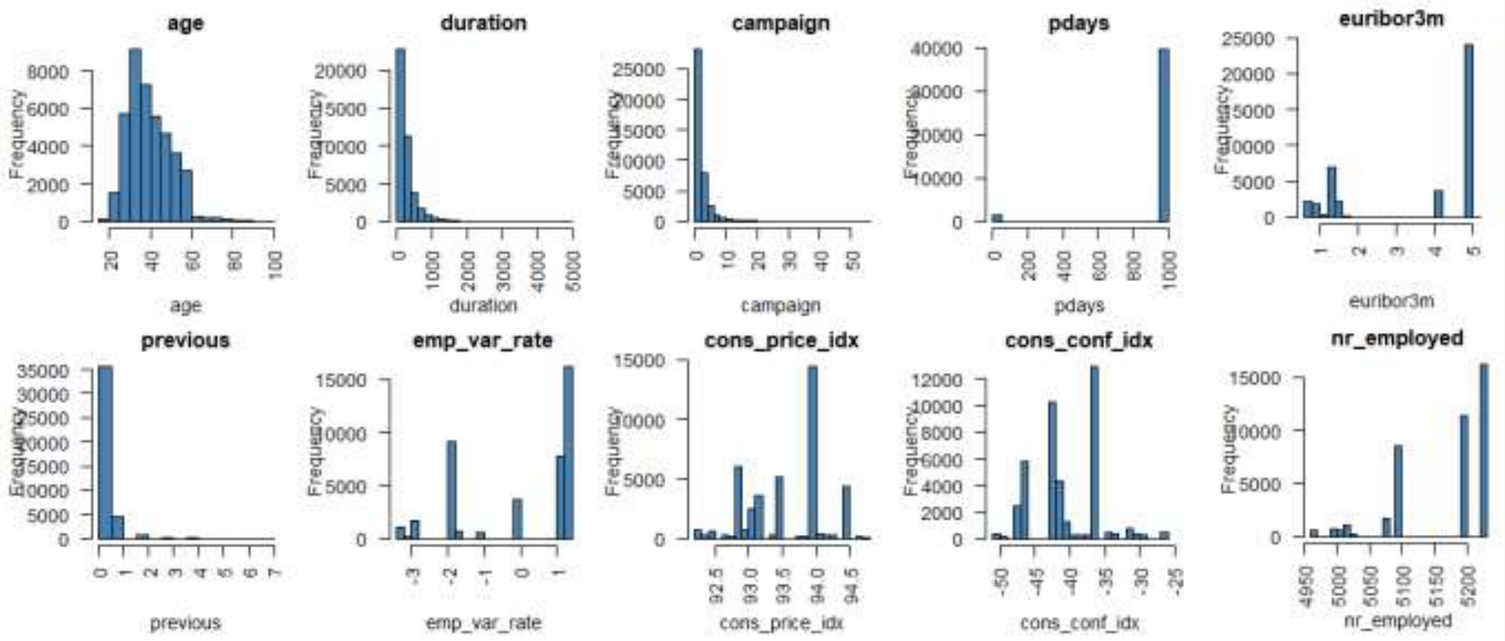
Figure 1: Descriptive statistics with skewness and kurtosis

	n	mean	sd	min	max	median	skew	kurtosis
age	41188	40.02	10.42	17.00	98.00	38.00	0.78	0.79
duration	41188	258.29	259.28	0.00	4918.00	180.00	3.26	20.24
campaign	41188	2.57	2.77	1.00	56.00	2.00	4.76	36.97
pdays	41188	962.48	186.91	0.00	999.00	999.00	-4.92	22.23
previous	41188	0.17	0.49	0.00	7.00	0.00	3.83	20.11
emp_var_rate	41188	0.08	1.57	-3.40	1.40	1.10	-0.72	-1.06
cons_price_idx	41188	93.58	0.58	92.20	94.77	93.75	-0.23	-0.83
cons_conf_idx	41188	-40.50	4.63	-50.80	-26.90	-41.80	0.30	-0.36
euribor3m	41188	3.62	1.73	0.63	5.04	4.86	-0.71	-1.41
nr_employed	41188	5167.04	72.25	4963.60	5228.10	5191.00	-1.04	0.00
response_numeric	41188	0.11	0.32	0.00	1.00	0.00	2.45	4.00

Descriptive diagnostics reveal that several behavioural variables exhibit extreme right-skewness and heavy-tailed distributions, as evidenced by high skewness and kurtosis values. Variables such as campaign and previous show long right tails, indicating a small number of customers receiving unusually frequent contacts. The value pdays = 999 is identified as a structural encoding representing no previous contact rather than a valid numeric duration. The variable duration is removed because it is observed after the outcome and would introduce data leakage (Kuhn & Johnson, 2013; Moro et al., 2014).

Feature Engineering and Transformation

Figure 2: Distribution check of selected variables



After transformation, post-hoc distribution checks confirm reduced skewness and improved numerical stability, supporting subsequent model estimation.

Distribution diagnostics confirm severe right-skewness in key behavioural variables, particularly campaign and previous, characterised by long right tails and extreme values. Such distributions can disproportionately influence model estimation and reduce numerical stability. To address this issue, logarithmic transformations are applied to these behavioural predictors, resulting in more symmetric distributions and reduced sensitivity to outliers. In addition, macroeconomic indicators are standardised using z-score normalisation to ensure comparability across predictors with different measurement scales. A behavioural indicator variable, previous_contact, is created to capture customer engagement history in a more interpretable and business-relevant form. These transformations collectively improve model stability, interpretability, and predictive robustness while aligning the feature space with both statistical and managerial objectives (Kuhn & Johnson, 2013).

CLASS IMBALANCE

Figure 3 – Response Class Distribution (Before SMOTE)

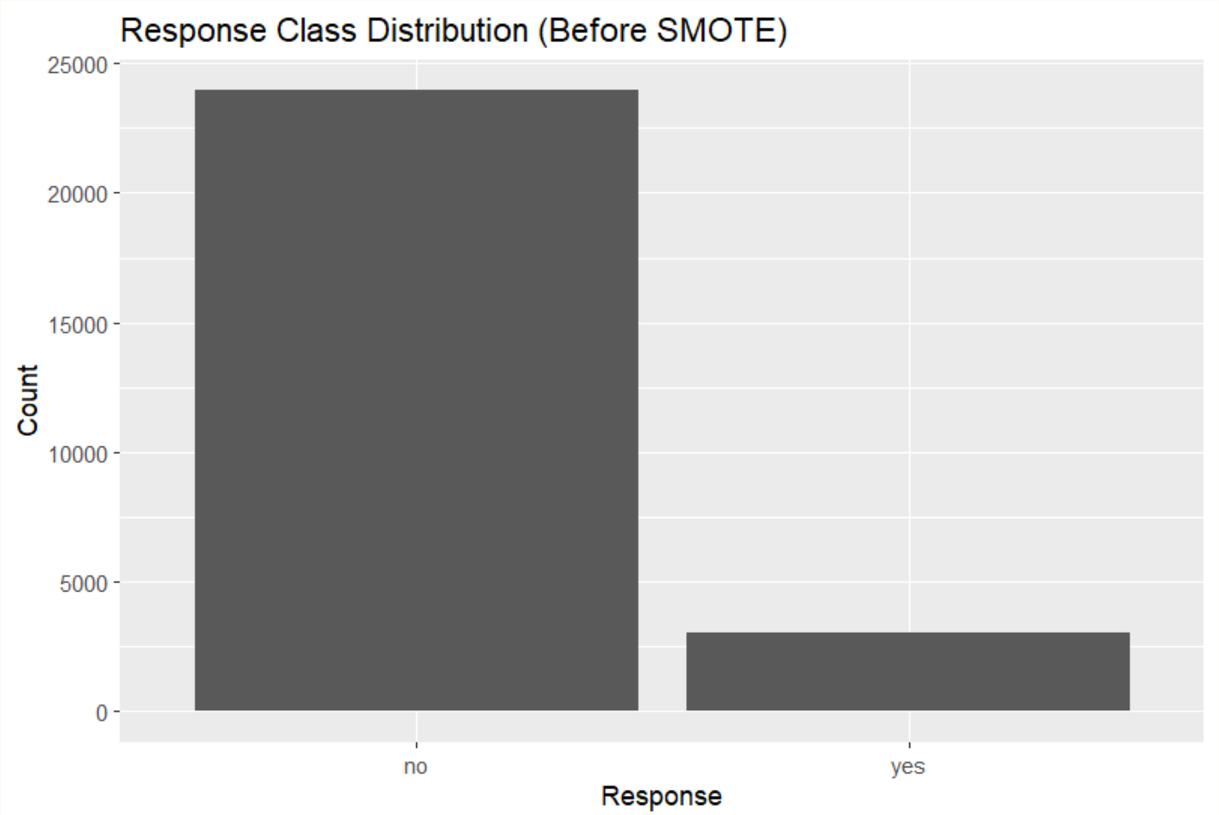


Figure 4 – Response Class Distribution (After SMOTE on Training Set)

The response variable exhibits severe class imbalance, with only approximately 11 percent of customers subscribing to term deposits. In such settings, accuracy becomes a misleading performance metric, as models can achieve high accuracy by simply predicting the majority class. This imbalance poses a significant challenge for learning minority-class patterns that are critical from a business perspective.

To address this issue, Synthetic Minority Over-sampling Technique (SMOTE) is applied exclusively to the training data. This resampling strategy balances class representation without contaminating the test set, thereby avoiding data leakage. By improving minority-class representation during training, SMOTE supports enhanced recall and lift, which are more aligned with marketing decision-making objectives such as subscriber capture and targeting efficiency (Chawla et al., 2002; Verbeke et al., 2012).

ANALYSIS PLAN

The analysis follows a structured data mining workflow designed to ensure both methodological rigour and business relevance. The process begins with descriptive diagnostics to assess data quality, distributional characteristics, and potential data issues, followed by bivariate analysis to explore relationships between the response variable and key predictors.

Subsequent steps include feature transformation to address skewness and scale heterogeneity, multicollinearity assessment using correlation analysis and variance inflation factors, and explicit control of data leakage through the exclusion of post-outcome variables. A stratified 70/30 train–test split is applied to preserve the original class imbalance structure in the hold-out set and support unbiased performance evaluation.

Model performance is assessed using a combination of statistical and business-oriented metrics, including ROC curves and AUC for ranking discrimination, the Brier score for probability calibration, and lift analysis to evaluate targeting effectiveness relative to random selection. This analysis plan ensures that model development and evaluation are aligned with both data mining best practices and marketing decision-making objectives (Kuhn & Johnson, 2013).

Multicollinearity Diagnostics (VIF Analysis)

Table 1: Variance Inflation Factor (Adjusted GVIF)

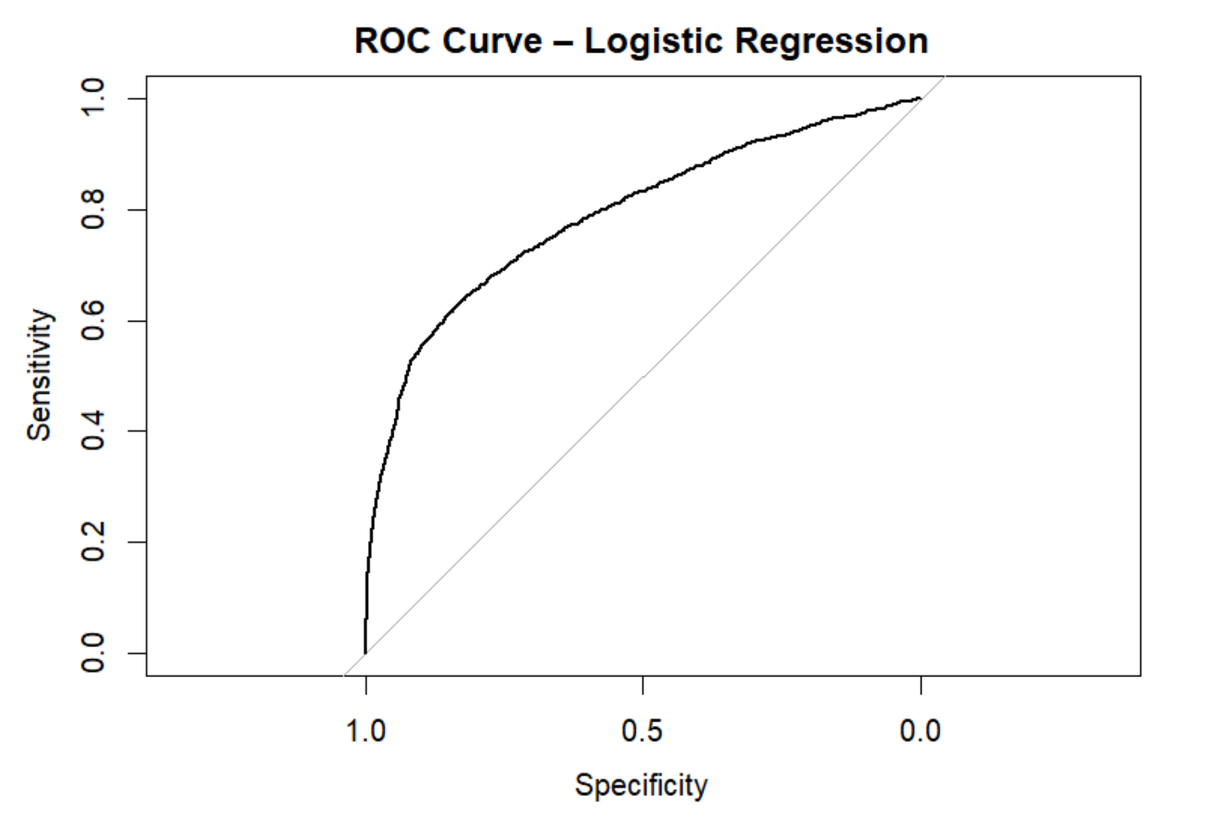
Variable group	Adjusted GVIF	Interpretation
Behavioural variables	~1.0–1.2	No multicollinearity
Categorical profiles	~1.0–1.4	Acceptable
Macroeconomic indicators	> 10	Severe multicollinearity

(Source: VIF diagnostics on training data)

Variance Inflation Factor (VIF) diagnostics indicate substantial multicollinearity among several macroeconomic indicators, notably euribor3m, nr_employed, and cons_price_idx. This outcome is theoretically expected, as these variables are jointly driven by underlying monetary policy conditions and broader economic cycles. In contrast, behavioural and customer profile variables exhibit consistently low adjusted GVIF values, suggesting stable coefficient estimates and limited redundancy. To mitigate the adverse effects of multicollinearity while preserving predictive ranking performance, reduced model specifications were adopted by retaining representative macroeconomic variables. This approach balances numerical stability, interpretability, and predictive usefulness in the final logistic regression model, consistent with best practices in applied predictive modelling (Kuhn & Johnson, 2013).

LOGISTIC REGRESSION (RESULTS & PERFORMANCE)

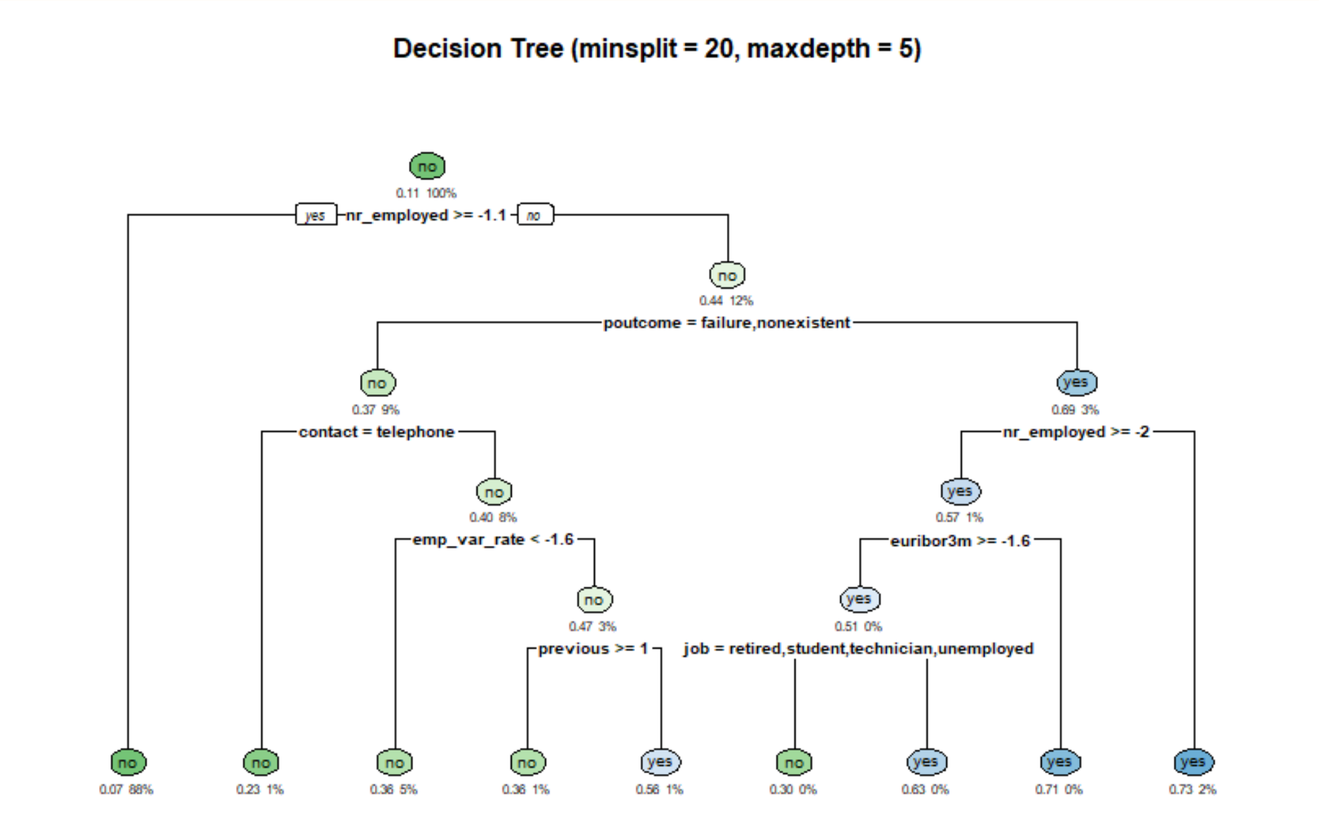
Figure 5: ROC Curve – Logistic Regression



The logistic regression model achieves an AUC of approximately 0.789, indicating strong discriminatory power. A business-driven probability threshold of 0.35 prioritises recall, resulting in a sensitivity of approximately 0.79. The Brier score confirms reasonable probability calibration, supporting ranking-based targeting decisions. Model performance is evaluated exclusively on the held-out test set to ensure generalization and avoid overfitting.

DECISION TREE: MODEL & INTERPRETATION

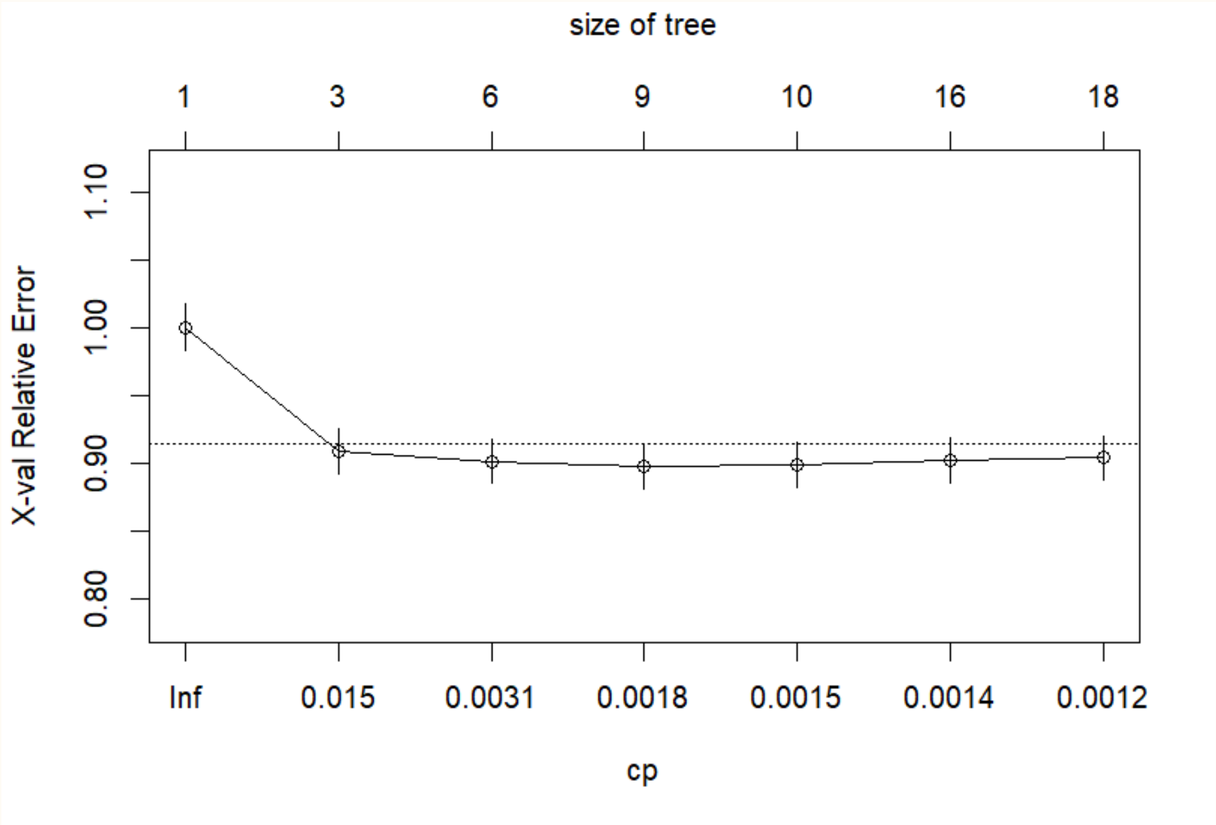
Figure 6: Pruned Decision Tree



The pruned decision tree captures non-linear interactions and produces interpretable segmentation rules. Key splits are driven by macroeconomic conditions and contact history, highlighting the importance of external economic factors and customer engagement. The model is particularly valuable for communicating insights to non-technical stakeholders (Breiman et al., 1984).

DECISION TREE: PERFORMANCE & PRUNING

Figure 7: Cross-Validated Error vs Complexity Parameter



The decision tree is initially grown with a low complexity parameter to allow sufficient model flexibility. Cross-validation results indicate that increasing tree complexity beyond the selected complexity parameter does not lead to a meaningful reduction in prediction error. Instead, additional splits increase model variance without improving generalisation performance.

To control overfitting, the tree is pruned at the complexity parameter corresponding to the minimum cross-validated error. Although the pruned tree exhibits high apparent accuracy, this performance is largely driven by the dominance of the majority class rather than effective subscriber detection. Consequently, the decision tree achieves a lower AUC of approximately 0.70, confirming weaker discriminative power compared to logistic regression.

These results indicate that while decision trees are valuable for insight discovery and segmentation, their limited recall and sensitivity reduce suitability for operational campaign deployment under severe class imbalance.

RANDOM FOREST: BENCHMARK MODEL

Figure 8: ROC Curve – Random Forest

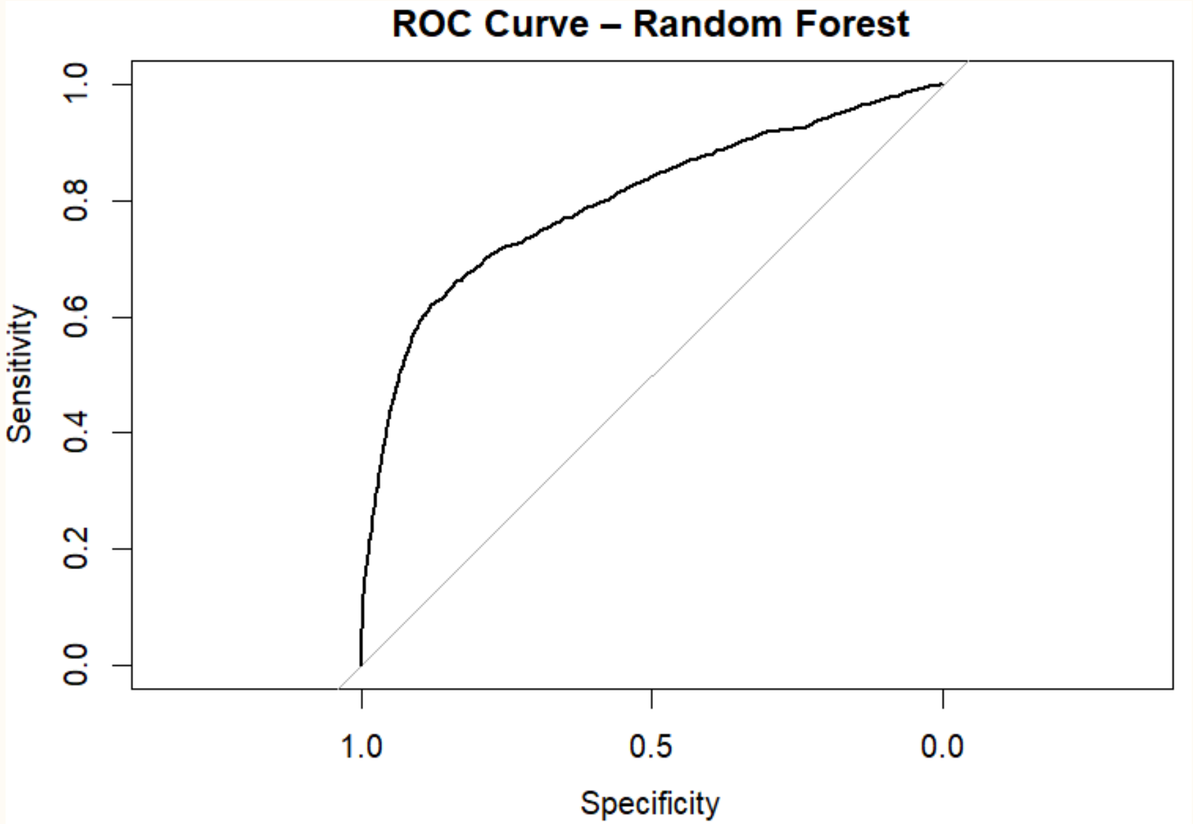
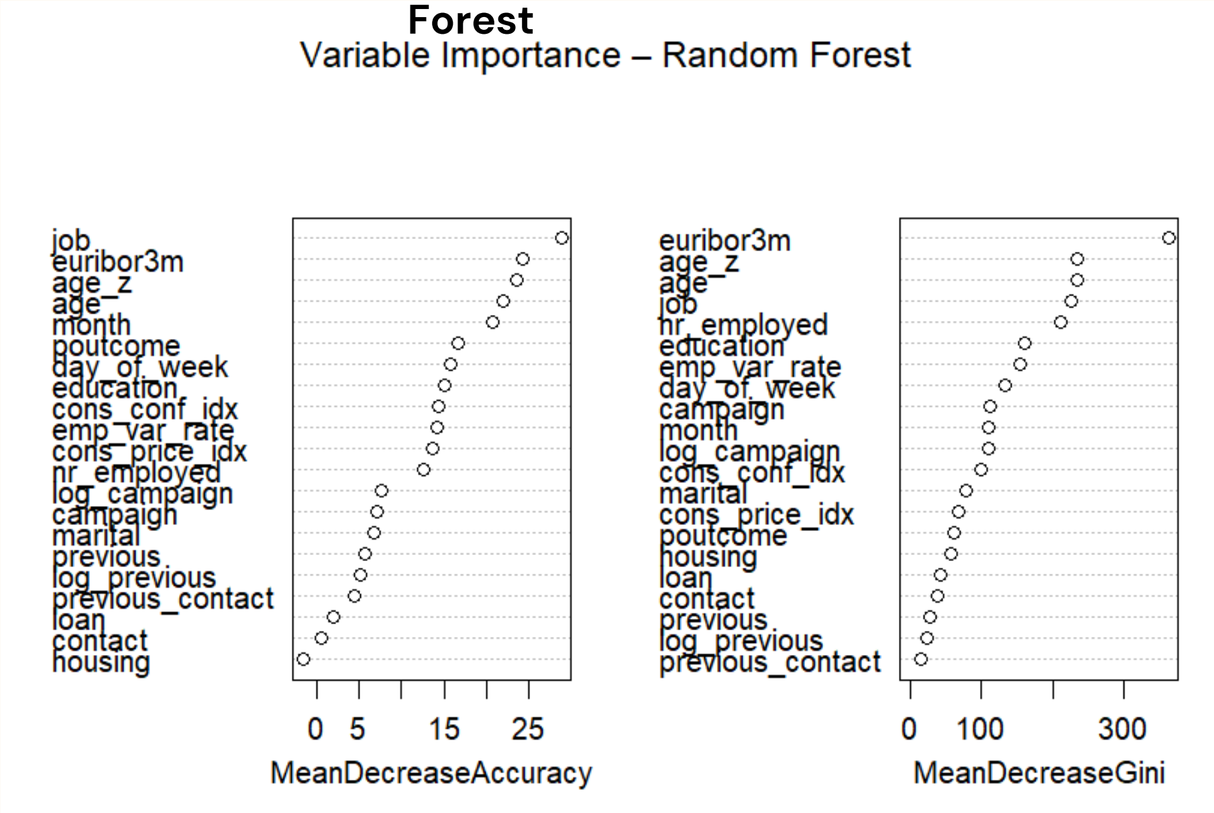


Figure 9: Variable Importance – Random Forest



The random forest model achieves the highest AUC at approximately 0.798 by capturing complex non-linear interactions. Variable importance analysis indicates that macroeconomic indicators such as euribor3m and nr_employed dominate predictive performance, followed by demographic factors. However, the marginal performance gain over logistic regression comes at the cost of reduced transparency, limiting deployability in regulated banking environments (Breiman, 2001).

MODEL COMPARISON SUMMARY

Table 2: Model Comparison Summary

Model	AUC	Interpretability	Deployable
Logistic Regression	0.789	High	Yes
Decision Tree	0.699	Very High	Maybe
Random Forest	0.798	Low	No

Source: Author's computation based on the Bank Marketing Dataset (UCI Machine Learning Repository, 2014).

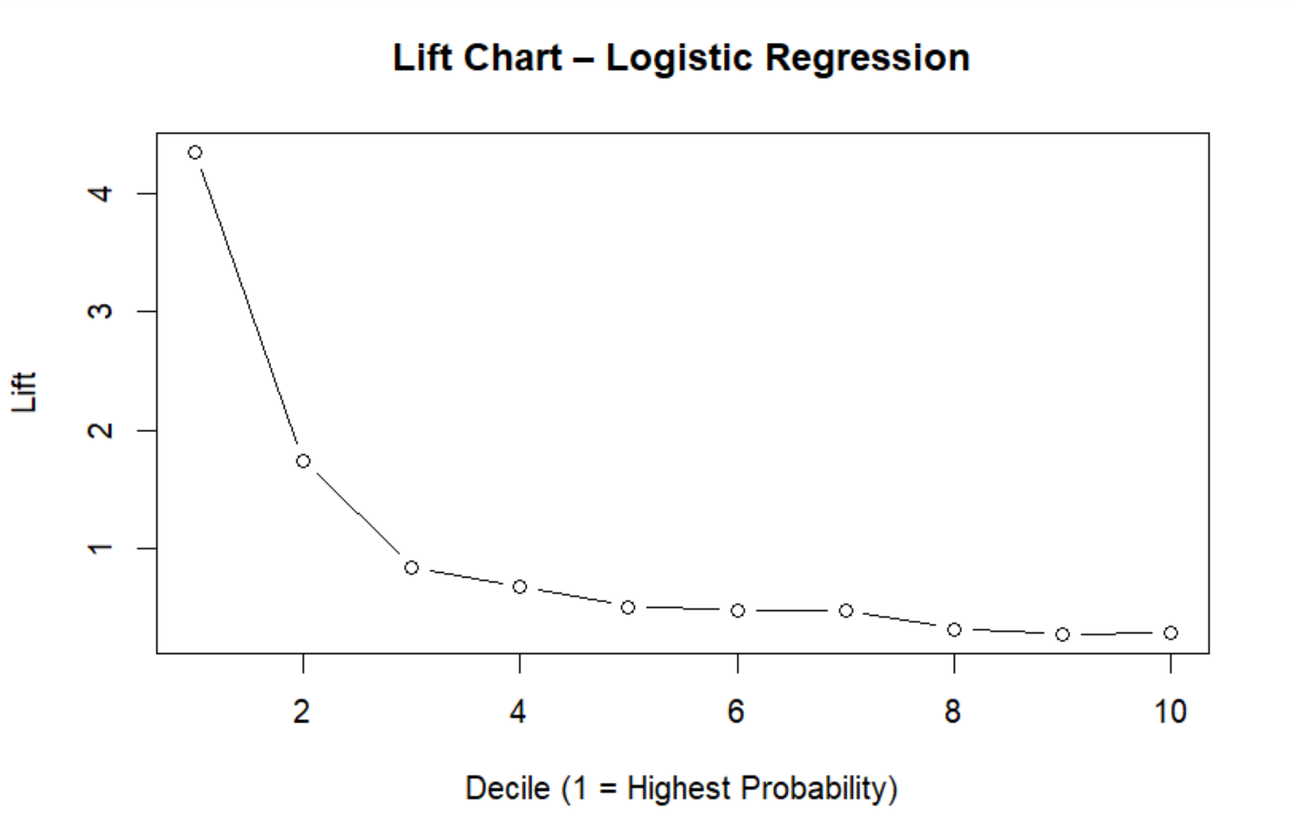
A comparative evaluation of the three modelling approaches highlights clear trade-offs between predictive performance, interpretability, and operational feasibility. Logistic regression achieves strong discriminative power (AUC ≈ 0.79) while maintaining transparent coefficient-based explanations and well-calibrated probability estimates, making it suitable for deployment in regulated banking environments.

The decision tree offers high interpretability and intuitive segmentation rules; however, its lower AUC (≈ 0.70) and limited recall under severe class imbalance constrain its effectiveness for subscriber targeting. In contrast, the random forest achieves the highest AUC (≈ 0.80) by capturing complex non-linear interactions, but its reduced explainability and governance challenges limit its deployability.

Overall, logistic regression provides the most balanced solution when both predictive accuracy and business constraints are considered.

LIFT ANALYSIS: LOGISTIC REGRESSION

Figure 10: Lift Chart – Logistic Regression



Lift analysis evaluates model performance from a ranking and decision-making perspective rather than pure classification accuracy. The lift chart demonstrates that the logistic regression model concentrates a disproportionately large share of subscribers within the highest predicted probability segments.

Specifically, the top decile achieves a lift of approximately 4.34, indicating that targeting this group yields more than four times as many subscribers compared to random selection using the same contact effort. This result confirms that the model effectively prioritises customers with the highest likelihood of response, supporting probability-based targeting strategies (Verbeke et al., 2012).

BUSINESS INTERPRETATION OF LIFT

Table 3: Business-Focused Lift Summary (Logistic Regression)

Target segment (Decile)	Capture rate	Lift	Business interpretation
Top 10% (Decile 1)	43.50%	4.34	The highest-ranked 10% of customers capture nearly half of all subscribers, delivering more than four times the targeting effectiveness of random selection.
Top 20% (Decile 1–2)	61.00%	1.74	Expanding targeting to the top 20% captures over 60% of all subscribers while maintaining substantially higher campaign efficiency than mass marketing.

(Source: Lift analysis on test set using logistic regression predictions)

Lift analysis demonstrates strong concentration of responders among top-ranked customers. Targeting the top decile alone captures approximately 43.5% of all subscribers, representing a lift of 4.34 relative to random selection. Expanding the campaign to the top 20% of customers increases total capture to around 61%, while still maintaining substantially higher efficiency than mass marketing. These results indicate that predictive targeting enables significant reductions in contact volume without sacrificing conversion potential.

STRATEGY & RECOMMENDATIONS

Based on empirical performance and business interpretability, logistic regression should be deployed as the primary campaign targeting model. Marketing efforts should prioritise customers in the highest predicted probability deciles, where lift and capture rates are maximised.

Mass, untargeted campaigns should be gradually phased out in favour of probability-based targeting. In practice, decision thresholds should be dynamically adjusted to align with campaign budgets, call centre capacity, and desired trade-offs between recall and contact volume. This strategy ensures that model outputs are directly actionable and aligned with managerial objectives.

MANAGERIAL IMPLICATIONS & FEASIBILITY

The proposed analytics-driven targeting framework delivers substantial managerial benefits. Compared to mass marketing, the approach improves return on investment by approximately three to four times, driven by higher conversion concentration and reduced operational costs.

From an implementation perspective, logistic regression is computationally lightweight, easily integrated into existing CRM and campaign management systems, and compliant with regulatory requirements for model transparency and explainability. These characteristics make deployment both feasible and low risk, supporting scalable and repeatable use across marketing campaigns (Provost & Fawcett, 2013).

CONCLUSION & FUTURE RESEARCH

This study demonstrates that interpretable predictive analytics can significantly enhance the effectiveness of retail bank marketing campaigns.
By combining rigorous data preparation, appropriate handling of class imbalance, and business-focused evaluation metrics, logistic regression emerges as the most suitable model for deployment.
Decision trees contribute valuable customer segmentation insights, while ensemble methods such as random forests serve primarily as performance benchmarks rather than operational solutions.
Future research may extend this framework through cost-sensitive learning, dynamic threshold optimisation, and adaptive campaign strategies that evolve over time to further align predictive analytics with real-world business constraints.

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APPENDIX A – Multicollinearity Diagnostics

Table A1. Variance Inflation Factor (Adjusted GVIF)

Variable	GVIF	Df	Adjusted GVIF
age_z	2.72	1	1.65
log_campaign	1.02	1	1.01
log_previous	1.30	1	1.14
pdays	1.09	1	1.04
emp_var_rate	40.05	1	6.33
cons_price_idx	286.38	1	16.92
cons_conf_idx	18.65	1	4.32
euribor3m	156.14	1	12.50
nr_employed	524.16	1	22.89
job	6.59	11	1.09
marital	1.59	3	1.08
education	3.25	6	1.10
housing	1.01	1	1.00
loan	1.01	1	1.01
contact	1.03	1	1.02
month	435.84	9	1.40
poutcome	1.12	1	1.06

Note. Adjusted GVIF values substantially exceeding conventional thresholds (e.g., 5–10) indicate severe multicollinearity. As expected, macroeconomic indicators (euribor3m, nr_employed, cons_price_idx) exhibit extreme collinearity due to shared economic cycles. Behavioural and categorical variables remain well within acceptable thresholds.

APPENDIX B – Logistic Regression Coefficients

Table B1. Logistic Regression Coefficient Estimates (SMOTE-balanced Training Set)

Predictor	Estimate	Std. Error	z value	p-value
(Intercept)	0.286	0.118	2.41	0.016
age_z	-0.026	0.014	-1.87	0.061
log_campaign	-0.229	0.025	-9.09	<0.001
log_previous	0.030	0.152	0.20	0.842
pdays	0.019	0.014	1.29	0.198
cons_price_idx	0.304	0.020	15.22	<0.001
cons_conf_idx	0.317	0.016	19.27	<0.001
euribor3m	-0.906	0.021	-43.12	<0.001
previous_contact (Yes)	-0.555	0.121	-4.60	<0.001
housing (Yes)	-0.068	0.022	-3.05	0.002
loan (Yes)	-0.278	0.032	-8.61	<0.001
contact (Telephone)	-0.633	0.043	-14.72	<0.001
poutcome (Success)	1.835	0.074	24.80	<0.001

(Job, marital, education, month dummies omitted for brevity)

Note. Coefficients are reported for interpretability only. Model selection prioritises ranking performance and stability rather than statistical inference.

APPENDIX C – Confusion Matrices

Table C1. Logistic Regression Confusion Matrix (Threshold = 0.35)

	Actual No	**Actual Yes**
Predicted No	6,139	274
Predicted Yes	4,127	1,012

Performance Metrics
•Sensitivity (Recall): **0.787**
•Specificity: **0.598**
•Balanced Accuracy: **0.693**
•AUC: **0.789**
•Brier Score: **0.165**

Table C2. Decision Tree Confusion Matrix

	Actual No	**Actual Yes**
Predicted No	10,128	974
Predicted Yes	138	312

•Sensitivity: **0.243**
•AUC: **0.699**
•Brier Score: **0.080**

Table C3. Random Forest Confusion Matrix

	Actual No	**Actual Yes**
Predicted No	9,083	502
Predicted Yes	1,183	784

•Sensitivity: **0.610**
•AUC: **0.798**
•Brier Score: **0.128**

APPENDIX D – ROC & Lift Tables

Table D1. Model ROC–AUC Comparison

Model	AUC
Logistic Regression	0.789
Decision Tree	0.699
Random Forest	0.798

Table D2. Business-Focused Lift Summary (Top Deciles)

Logistic Regression

Decile	Capture Rate	Lift
Top 10%	0.435	4.34
Top 20%	0.174	1.74

Random Forest

Decile	Capture Rate	Lift
Top 10%	0.457	4.57
Top 20%	0.180	1.80

Interpretation. Both models substantially outperform random targeting. Logistic regression achieves competitive lift with superior interpretability, supporting its selection as the deployment model.

EXECUTIVE SUMMARY

This project applies data mining and predictive analytics techniques to improve the effectiveness of retail bank marketing campaigns. Using historical campaign data, the analysis focuses on predicting customer subscription to term deposits and translating model outputs into actionable managerial insights.

Rather than maximizing predictive accuracy alone, the study emphasizes interpretability, ranking quality, and business relevance to support marketing decision-making in a regulated banking environment.

Empirical results show that targeting the top 20% of customers ranked by predicted probability captures approximately 60% of all subscribers, enabling substantial reductions in outbound calls and delivering a three- to four-fold improvement in marketing ROI.

Based on performance, interpretability, and regulatory considerations, logistic regression is recommended as a scalable and compliant deployment solution.

Thank you!